Two-dimensional column vectors provide another concrete example. We construct them by stacking up a pair of complex numbers, $\alpha_1$ and $\alpha_2$, in the form

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

and identifying this "stack" with the ket-vector $|A\rangle$. The complex numbers $\alpha$ are the components of $|A\rangle$. You can add two column vectors by adding their components:

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 + \beta_1 \\ \alpha_2 + \beta_2 \end{pmatrix}.$$

Moreover, you can multiply the column vector by a complex number $z$ just by multiplying the components,

$$z \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} z\alpha_1 \\ z\alpha_2 \end{pmatrix}.$$

Column vector spaces of any number of dimensions can be constructed. For example, here is a five-dimensional column vector:

$$\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{pmatrix}.$$

Normally, we do not mix vectors of different dimensionality.